

DBMS – PRODUCT AND CATEGORY MANAGEMENT

Project Title: ONE8 Footwear E-Commerce System

Skill Task: DBMS

Week: 2

1. Database Design

1.1 Category Table

Attribute	Data Type	Constraint
Category_ID	NUMBER	Primary Key
Category_Name	VARCHAR2(50)	NOT NULL
Description	VARCHAR2(255)	—

1.2 Product Table

Attribute	Data Type	Constraint
Product_ID	NUMBER	Primary Key
Product_Name	VARCHAR2(100)	NOT NULL
Brand_ID	NUMBER	—
Category_ID	NUMBER	NOT NULL, Foreign Key
Price	NUMBER(10,2)	NOT NULL
Size	VARCHAR2(10)	—
Color	VARCHAR2(50)	—
Stock	NUMBER	NOT NULL

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship: One-to-Many (1:M)

3. Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID      NUMBER PRIMARY KEY,  
3     Category_Name    VARCHAR2(50) NOT NULL,  
4     Description       VARCHAR2(255)  
5 );
```

Table created.

3.2 Create Product Table

```
SQL> CREATE TABLE Product (  
2     Product_ID       NUMBER PRIMARY KEY,  
3     Product_Name      VARCHAR2(100) NOT NULL,  
4     Brand_ID          NUMBER,  
5     Category_ID       NUMBER NOT NULL,  
6     Price             NUMBER(10,2) NOT NULL,  
7     Size              VARCHAR2(10),  
8     Color              VARCHAR2(50),  
9     Stock              NUMBER NOT NULL,  
10    CONSTRAINT FK_Product_Category  
11         FOREIGN KEY (Category_ID)  
12         REFERENCES Category(Category_ID)  
13 );
```

Table created.

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)
      2 VALUES (1, 'Running', 'Running shoes for daily training and sports');
```

1 row created.

SQL>

```
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)
      2 VALUES (2, 'Casual', 'Everyday casual footwear');
```

1 row created.

SQL>

```
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)
      2 VALUES (3, 'Formal', 'Formal and office wear shoes');
```

1 row created.

SQL>

```
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)
      2 VALUES (4, 'Sneakers', 'Lifestyle and streetwear sneakers');
```

1 row created.

SQL>

```
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)
      2 VALUES (5, 'Sports', 'Performance footwear for outdoor sports');
```

1 row created.

4.2 Product Records

```
SQL> INSERT INTO Product
  2 VALUES (101, 'Air Max Runner', 1, 1, 3499.00, '9', 'Black', 40);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (102, 'Cloud Stride', 2, 1, 3999.00, '8', 'Blue', 35);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (103, 'Urban Casual Slip-On', 3, 2, 1999.00, '9', 'Grey', 50);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (104, 'Everyday Canvas', 4, 2, 1499.00, '8', 'White', 45);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (105, 'Oxford Classic', 5, 3, 4499.00, '10', 'Black', 20);

1 row created.
```

```
SQL> INSERT INTO Product
  2 VALUES (106, 'Derby Elite', 6, 3, 4999.00, '9', 'Brown', 18);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (107, 'Street Icon Sneaker', 7, 4, 5499.00, '9', 'White', 30);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (108, 'Retro Bold', 8, 4, 5999.00, '10', 'Red', 25);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (109, 'Trail Blazer Pro', 9, 5, 4299.00, '9', 'Black', 28);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (110, 'Court Smash', 10, 5, 3799.00, '8', 'Green', 32);

1 row created.
```

5. Data Retrieval

5.1 Display Category Records

```
SQL> SELECT * FROM Category;

CATEGORY_ID CATEGORY_NAME
-----
DESCRIPTION
-----
          1 Running
Running shoes for daily training and sports

          2 Casual
Everyday casual footwear

          3 Formal
Formal and office wear shoes

          4 Sneakers
Lifestyle and streetwear sneakers

          5 Sports
Performance footwear for outdoor sports

5 rows selected.
```

5.2 Display Product Records

```
SQL> SELECT * FROM Product;

PRODUCT_ID PRODUCT_NAME          BRAND_ID CATEGORY_ID    PRICE SIZE  COLOR  STOCK
-----
          101 Air Max Runner          1          1    3499.00  9    Black   40
          102 Cloud Stride            2          1    3999.00  8    Blue    35
          103 Urban Casual Slip-On     3          2    1999.00  9    Grey    50
          104 Everyday Canvas          4          2    1499.00  8    White   45
          105 Oxford Classic           5          3    4499.00 10    Black   20
          106 Derby Elite              6          3    4999.00  9    Brown   18
          107 Street Icon Sneaker      7          4    5499.00  9    White   30
          108 Retro Bold               8          4    5999.00 10    Red     25
          109 Trail Blazer Pro         9          5    4299.00  9    Black   28
          110 Court Smash              10         5    3799.00  8    Green   32

10 rows selected.
```

6. Product Update

Update the price and stock of Air Max Runner:

```
SQL> UPDATE Product
  2  SET Price = 3299.00,
  3      Stock = 45
  4  WHERE Product_ID = 101;
```

```
1 row updated.
```

Update the price of Cloud Stride:

```
SQL> UPDATE Product
  2  SET Price = 4199.00
  3  WHERE Product_Name = 'Cloud Stride';
```

```
1 row updated.
```

7. Product Deletion

Delete the product Court Smash:

```
SQL> DELETE FROM Product
  2  WHERE Product_ID = 110;
```

```
1 row deleted.
```

Delete the product Trail Blazer Pro:

```
SQL> DELETE FROM Product
  2  WHERE Product_Name = 'Trail Blazer Pro';
```

```
1 row deleted.
```

8. Category-wise Product Reports

8.1 Product Listing by Category

This report displays the category name along with the products belonging to each category, including their price and stock.

```

SQL> SELECT
  2     c.Category_Name,
  3     p.Product_ID,
  4     p.Product_Name,
  5     p.Price,
  6     p.Stock
  7 FROM Category c
  8 JOIN Product p
  9     ON c.Category_ID = p.Category_ID
 10 ORDER BY c.Category_Name, p.Product_Name;

```

CATEGORY_NAME	PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK
Casual	103	Urban Casual Slip-On	1999.00	50
Casual	104	Everyday Canvas	1499.00	45
Formal	105	Oxford Classic	4499.00	20
Formal	106	Derby Elite	4999.00	18
Running	101	Air Max Runner	3299.00	45
Running	102	Cloud Stride	4199.00	35
Sneakers	107	Street Icon Sneaker	5499.00	30
Sneakers	108	Retro Bold	5999.00	25

8 rows selected.

8.2 Products in Sneakers Category

This report displays the products belonging specifically to the Sneakers category.

```

SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     p.Price,
  5     p.Stock
  6 FROM Product p
  7 JOIN Category c
  8     ON p.Category_ID = c.Category_ID
  9 WHERE c.Category_Name = 'Sneakers';

```

PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK
107	Street Icon Sneaker	5499.00	30
108	Retro Bold	5999.00	25

2 rows selected.

8.3 Product Count by Category

This report displays the number of products available in each category.

```
SQL> SELECT
  2     c.Category_Name,
  3     COUNT(p.Product_ID) AS Total
  4 FROM Category c
  5 LEFT JOIN Product p
  6     ON c.Category_ID = p.Category_ID
  7 GROUP BY c.Category_Name
  8 ORDER BY c.Category_Name;
```

CATEGORY_NAME	TOTAL_PRODUCTS
Casual	2
Formal	2
Running	2
Sneakers	2
Sports	0

5 rows selected.